Recurring Decimals

Exam code: 0607

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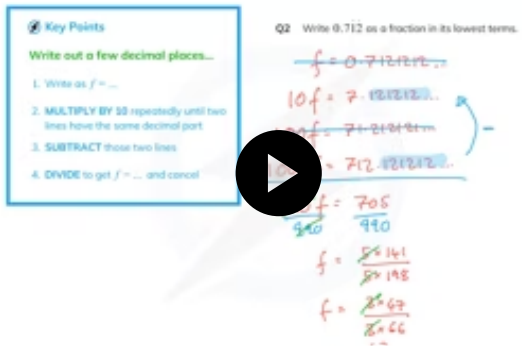

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Recurring decimals

What are recurring decimals?

- When writing a **rational number** as a decimal, it will either be:
  - A decimal that stops, called a "**terminating**" decimal
    - $\frac{1}{4} = 0.25$
  - Or a decimal that repeats with a pattern, called a "**recurring**" decimal
    - $\frac{32}{99} = 0.32323232...$
- The recurring part can be written with a **dot above the digit that repeats**
- If multiple digits repeat, dots are used on the **first and last digits that repeat**
  - $0.3333... = 0.\dot{3}$
  - $0.121212... = 0.\dot{1}\dot{2}$
  - $0.325632563256... = 0.\dot{3}25\dot{6}$
- **Recurring decimals are rational numbers**, they are not irrational
  - They can be **written as fractions**

How do I write recurring decimals as fractions?

Help me understand

Stuck on something?

Ask questions and get help with exactly what you're revising.

Got it

- STEP 1
  Write the recurring decimal as $x = ...$
  - $x = 0.35353535...$

- STEP 2
  Multiply both sides by 10 repeatedly until two lines have the same recurring decimal part
  - $x = 0.35353535...$
  - $10x = 3.5353535...$
  - $100x = 35.353535...$
    - Note that x and $100x$ have 35 repeating after the decimal point, the repeating pattern after $10x$ is 53 repeating

- STEP 3
  Subtract the two lines which have matching recurring decimal parts
  - $100x - x = 35.353535... - 0.35353535...$
  - $99x = 35$

- STEP 4
  Divide both sides to get $x = ...$
  Cancel if necessary to get fraction in its lowest terms
  - $x = \frac{35}{99}$

Worked Example

Write $0.\dot{3}0\dot{7}$ as a fraction in its lowest terms.

Answer:

Write as $x = ...$ to show the pattern

$$x = 0.307307307307...$$

Multiply both sides by 10 repeatedly until two lines have the same recurring decimal part

$$10x = 3.07307307307...$$
$$100x = 30.730730730...$$
$$1000x = 307.307307307...$$

Notice that x and $1000x$ have matching recurring decimal parts

Subtract one from the other

$$1000x - x = 307.307307307... - 0.307307307...$$
$$999x = 307$$

Divide both sides by 999

$$x = \frac{307}{999}$$

This cannot be simplified, so this is the final answer

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